

RM A. Villa

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SUMMARY

Computer Science undergraduate with a dual focus on scalable backend architecture and intelligent system design. Experienced in bridging software engineering fundamentals with applied machine learning, from building high-throughput APIs to developing data-driven predictive models. Passionate about solving complex research and engineering problems using Python, SQL, and modern ML pipelines/frameworks.

EDUCATION

Manuel S. Enverga University Foundation
B.S. in Computer Science

2024 – Present
Lucena City, Philippines

HACKATHONS & LEADERSHIP

Philippine Junior Data Science Challenge (PJDSC 2025)

2025

Backend Developer, Top 10 National Finalist [GitHub]

Manila, Philippines

- Engineered the **FastAPI & SQLAlchemy** backend for **PRESKO**, a climate-resilience web app providing real-time heat-risk monitoring and personalized safety recommendations to citizens.
- Architected a hybrid heat-data ingestion pipeline leveraging **APScheduler** and **Open-Meteo API**, implementing data fetching and a 1-hour caching strategy to ensure minimal API overhead and low-latency responses.
- Integrated **Cohere LLM** to generate personalized, heat-adaptation strategies mapped to polymorphic user profiles (e.g., outdoor workers, students), utilizing a 180-minute caching layer to reduce AI inference costs.
- Implemented geospatial endpoints using **Haversine distance algorithms** to match users with real-time nearest cooling/hydration centers, backed by SQLite/PostgreSQL and local storage fallbacks.

PROJECTS

MajorMatch [GitHub] | Python, Streamlit, PostgreSQL, Sentence-Transformers, scikit-learn, UMAP

2026

- Engineered a semantic course discovery and career pathfinder system using **Sentence-Transformers** and **PostgreSQL** to deliver vector-based course recommendations and track predictions.
- Implemented custom **Cosine Similarity** vector search routines alongside automated data pipelines for embedding generation, batch ingestion, and database indexing via **SQLAlchemy**.
- Designed 2D interactive embedding visualizations using **PCA, UMAP, and t-SNE** paired with **Plotly**, allowing users to visually map academic offerings to career trajectories.
- Architected a **Streamlit** UI with tool-calling orchestration to seamlessly integrate semantic course retrieval, career context analytics, and real-time model inference.

BarangayMicroJobs [GitHub] | Flutter, FastAPI, PostgreSQL, SQLAlchemy, JWT, Docker

2025

- Architected a full-stack hyper-local micro-employment platform with a **Flutter** mobile frontend and a high-performance **FastAPI** REST backend.
- Engineered transactional backend endpoints using **SQLModel** and **PostgreSQL** to securely process job postings, candidate applications, escrow transactions, and mutual ratings.
- Implemented secure JWT authentication and password hashing utilizing **bcrypt, passlib, and python-jose**, enforcing CORS policy and role-based route protection.
- Developed offline-first data synchronization in Flutter via an **OfflineSyncService** to ensure resilient mobile UX in low-connectivity environments.

LIGTAS | React, FastAPI, SQLAlchemy, Leaflet

2025

- Developed a real-time earthquake monitoring and safety guidance platform tailored for Filipino users.
- Built a mobile-first dashboard with earthquake feeds, safety guides, and an evacuation-center map.
- Integrated a FastAPI backend and interactive mapping features for public-safety awareness and response.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, JS/HTML/CSS

Frameworks: FastAPI, Django, Flutter, React

Tools: Git, GitHub, Docker, Redis, Pytest, dotenv, Streamlit, AI

Databases: PostgreSQL, SQLite

ML/AI: Scikit-learn, Pandas, NumPy, Matplotlib, RAG, Embeddings, Semantic Search, Tool Calling